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Experiment 4: Binary Classification using Linear and Kernel-Based Models

Objective

To classify emails as spam or ham using Logistic Regression and Support Vector Machine (SVM) classifiers and to analyze the effect of hyperparameter tuning on classification performance.

Dataset

The **Spambase** dataset contains numerical features extracted from email content and a binary label indicating spam or non-spam (ham).

Dataset Links (for reference):

- Kaggle: https://www.kaggle.com/datasets/somesh24/spambase

3. Preprocessing Steps

The following preprocessing steps were applied to prepare the dataset for effective model training and evaluation.

3.1 Missing Value Check

The dataset was examined for missing or null values. No missing values were detected, and therefore no imputation was required.

3.2 Feature Standardization

All numerical features were standardized using `StandardScaler` to achieve a mean of 0 and a standard deviation of 1. Feature scaling is essential for models such as Support Vector Machines (SVM) and Logistic Regression, as these algorithms are sensitive to the relative scale of input features.

3.3 Train–Test Split

The dataset was partitioned into training and testing subsets using an 80:20 split. The training set was used to learn model parameters, while the testing set was reserved for evaluating model generalization on unseen data.

4. Implementation Details

The models were implemented using the `scikit-learn` library with the following configurations and tuning strategies.

4.1 Logistic Regression

Logistic Regression models were trained using multiple solvers, including `liblinear` and `saga`. Both $L1$ (Lasso) and $L2$ (Ridge) regularization techniques were evaluated. The inverse regularization strength parameter C was tuned over a predefined range to identify the optimal balance between bias and variance.

4.2 Support Vector Machine (SVM)

Support Vector Machine classifiers were evaluated using four different kernel functions: Linear, Polynomial, Radial Basis Function (RBF), and Sigmoid. Hyperparameters tuned during model selection included the regularization parameter C (ranging from 0.1 to 100), the kernel coefficient `gamma` (with values `scale` and `auto`), and the polynomial degree for polynomial kernels.

4.3 Validation Strategy

A 5-Fold Cross-Validation strategy was employed during hyperparameter tuning to ensure stability and robustness of the experimental results.

5. Visualizations

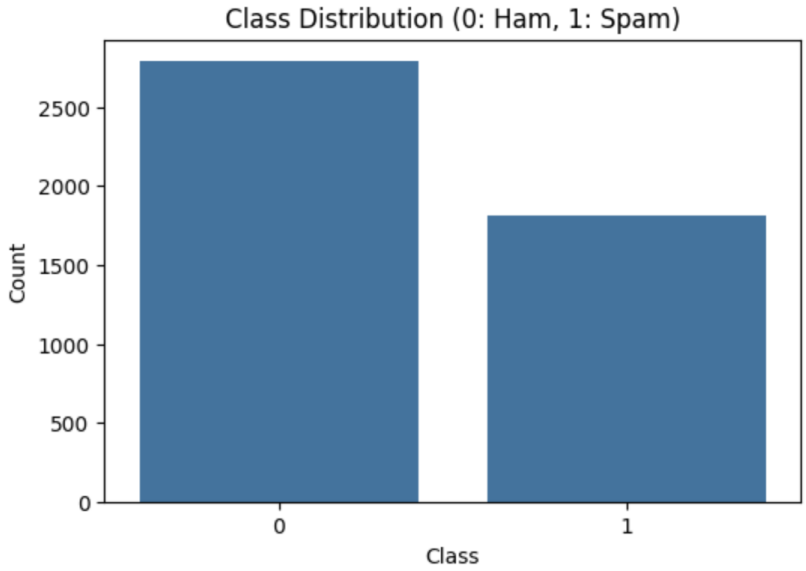

Figure 1: class Distribution

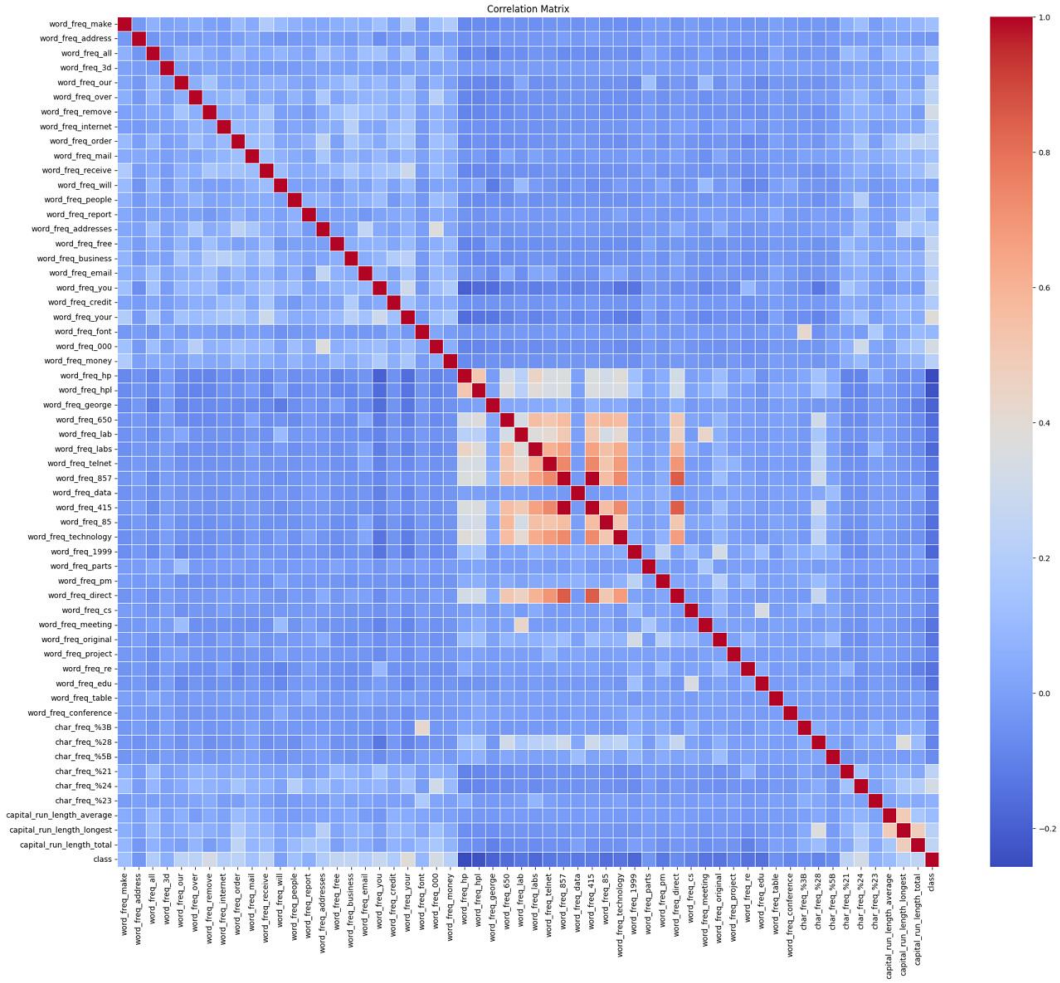

Figure 2: Correlation

Hyperparameter Tuning Results

Model	Search Method	Best Parameters	Best CV Accuracy
Logistic Regression	Grid	C=100, Penalty=L2, Solver=liblinear	0.919
SVM	Grid	C=0.1, Gamma=scale, Kernel=sigmoid	0.651

Metric	Value
Accuracy	0.939
Precision	0.939
Recall	0.917
F1 Score	0.917
Training Time (s)	0.039

Kernel	Accuracy	F1 Score	Training Time (s)
Polynomial	0.674	0.253	1.030
RBF	0.720	0.569	0.391
Sigmoid	0.652	0.551	0.303

Fold	Logistic Regression	SVM
Fold 1	0.917	0.655
Fold 2	0.922	0.731
Fold 3	0.888	0.698
Fold 4	0.940	0.729
Fold 5	0.831	0.709
Average	0.900	0.705

Criterion	Logistic Regression	SVM
Accuracy	93.9%	70.5%
Model Complexity	Low	High
Training Time	Low	High
Interpretability	High	Low

Logistic Regression Performance

SVM Kernel-wise Performance

K-Fold Cross-Validation Results (K = 5)

Comparative Analysis

Performance Analysis and Discussion

The experimental results reveal a significant performance disparity between the Logistic Regression and Support Vector Machine (SVM) models when applied to the Spambase dataset.

1. **Comparative Model Performance**

   - **Best Overall Classifier:** Logistic Regression emerged as the superior model, achieving a high accuracy of **0.939** and an F1-score of **0.917**. This significantly outperformed the best SVM kernel (RBF), which reached an accuracy of only 0.720.
   - **Kernel Comparison:** Within the SVM trials, the **RBF kernel** was the most effective (0.720 accuracy), followed by Polynomial (0.674) and Sigmoid (0.652). The higher performance of the RBF kernel suggests the presence of non-linear relationships in the spam data that simpler kernels struggled to capture.
   - **Computational Efficiency:** Logistic Regression was notably more efficient, with a training time of **0.039s**, compared to the SVM kernels which required between 0.303s and 1.030s.

2. **Impact of Hyperparameter Tuning**

   - **Logistic Regression Tuning:** Grid Search identified the optimal configuration as **C=100 with L2 penalty**. The high C value indicates that weak regularization was necessary to achieve high predictive power, allowing the model to fit the training data more closely to identify relevant spam patterns.
   - **SVM Tuning:** The best SVM configuration found was **C=0.1 with a Sigmoid kernel** (0.651 CV accuracy). The low C value suggests that the SVM required strong regularization to maintain a stable margin, as higher C values likely led to overfitting.

3. **Bias–Variance and Generalization**

   - **Generalization:** Logistic Regression demonstrated excellent stability, with an average 5-fold cross-validation score of **0.900**. The consistency between its test accuracy (0.939) and CV average indicates low variance and strong generalization.
   - **Underfitting in SVM:** The SVM model exhibited signs of **high bias**, particularly with the Sigmoid and Polynomial kernels, resulting in an average CV score of **0.705**. This suggests the SVM models were too simple to fully capture the underlying data distribution, leading to underfitting.

4. **Summary**

The margin-based optimization of SVM proved less effective for this high-dimensional numerical dataset than the probabilistic approach of Logistic Regression. While the RBF kernel improved SVM's ability to handle non-linearity, Logistic Regression's balanced approach to bias and variance provided the most robust results for identifying spam.

Learning Outcomes

- Understand probabilistic and margin-based classifiers.
- Apply hyperparameter tuning.
- Evaluate classification models.
- Interpret experimental results.

References

- Scikit-learn: Logistic Regression
- Scikit-learn: Support Vector Machines
- Scikit-learn: Hyperparameter Optimization
- Spambase Dataset – Kaggle
- UCI ML Repository – Spambase